

04/20/2026

Toward full faithfulness

of KZ

Last time Descent for idempotent algebras, started the proof that for (N, Σ) smooth projective,

$$R_{\Sigma} \lim_{\sigma \in \Sigma^{\text{op}}} \text{Mod}_{\omega_{\sigma, \nu}} \text{Sh}(M_R) \longrightarrow \text{Sh}(M_R)$$

is an equivalence.

Today. Complete the proof of the above, start the proof that for Σ smooth, k_{Σ} is fully faithful.

Correction from last time. We said that for $e \in \text{CAlg}(\text{Pr}^L)$, the poset $\text{CAlg}^{\text{idem}}(e)$ has all meets, we meant to say finite meets.

Improvement from last time. For R_{Σ} to be an equivalence, we only need Σ to be projective (smoothness is not needed). We'll recall this result presently.

Finishing the proof of the gluing formula for the idempotent algebras ω_σ

> We started the proof of

Thm. Let (N, Σ) be a projective fan. Then the natural maps

$$\mathbb{1}_{\text{conv}} \longrightarrow \lim_{\sigma \in \Sigma^{\text{op}}} \omega_\sigma \longrightarrow \lim_{S \in \text{Sub}_{\neq \emptyset}(\Sigma_{\text{top}})} \bigstar_{\tau \in S} \omega_\tau$$

are equivalences of idempotent algebras in $(\text{Sh}(M_{\mathbb{R}}), *)$

> We saw that the right-hand map is always an equivalence.

> Recall that $\mathbb{1}_{\text{conv}} = \mathcal{O}_*(S)$ is the skyscraper sheaf

> We checked that the left-hand map is an equiv. on the stalk at 0. So we need to see the stalks of the middle term at $m \in M_{\mathbb{R}} - \{0\}$ are zero.

Stalk at $m \in M_{\mathbb{R}} \setminus \{0\}$.

We want to show that the stalk of the limit is zero

we see that

$$\lim_{\sigma \in \Sigma^{\text{op}}} (\omega_{\sigma, \nu})_m \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \mathcal{S}_{m, +}[\eta],$$

where

$$\begin{aligned} \mathcal{S}_{m, +} : \Sigma^{\text{op}} &\longrightarrow \mathcal{S}_p \\ \sigma &\longmapsto \begin{cases} \mathcal{S}, & m \in \sigma^{\vee, 0} \\ 0, & \text{else} \end{cases} \end{aligned}$$

Write $\Sigma_{m, -} = \{\sigma \notin \Sigma \mid m \in \sigma^{\vee, 0}\}$, and $\mathcal{S}_{m, -}$ for the right Kan extension of the constant presheaf $\Sigma_{m, -}^{\text{op}} \rightarrow \mathcal{S}_p$ at $\mathcal{S}$ to Σ^{op} . Then there is a fiber sequence

$$\mathcal{S}_{m, +} \longrightarrow \underline{\mathcal{S}} \longrightarrow \mathcal{S}_{m, -}$$

in $\text{Fun}(\Sigma^{\text{op}}, \mathcal{S}_p)$. Thus to prove the claim, it suffices to

Show that the right-hand map induces an isomorphism on limits. As before, since $S_{m,-}$ is right Kan extended from $\Sigma_{m,-}^{\text{op}}$,

$$\lim_{\sigma \in \Sigma^{\text{op}}} S_{m,-} \simeq \lim_{\sigma \in \Sigma_{m,-}^{\text{op}}} \underline{S}$$

Moreover, the map

$$\underline{S} \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \underline{S} \longrightarrow \lim_{\sigma \in \Sigma_{m,-}^{\text{op}}} \underline{S} \simeq \lim_{B(\Sigma_{m,-}^{\text{op}})} \underline{S}$$

$\nearrow$
 Σ^{op} is contractible

is induced by the inclusion $\Sigma_{m,-}^{\text{op}} \subset \Sigma^{\text{op}}$. Thus we are reduced to showing that the poset $\Sigma_{m,-}$ has contractible classifying anima.

Weak contractibility of $\Sigma_{m,-}^{\text{op}}$

Here, we use the projectivity of Σ . Fix a moment polytope

$P \subset M_{\mathbb{R}}$ for Σ , so that Σ is the normal fan Σ_P of P .

Then there is a canonical isomorphism of posets

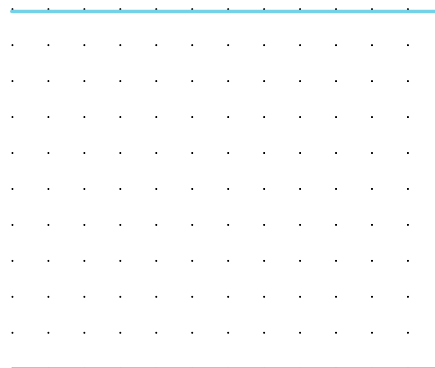
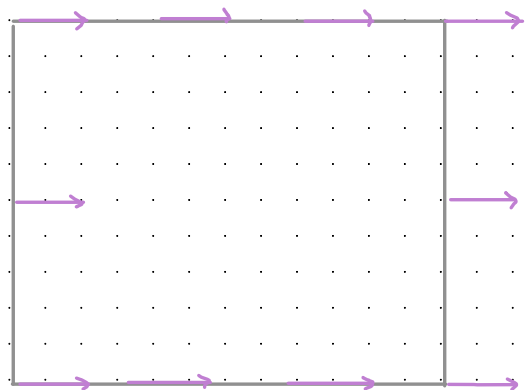
$$\text{Face}(P) \longrightarrow \Sigma^{\text{op}}$$

$$Q \longmapsto \sigma_Q = \{v \in N_{\mathbb{R}} \mid \forall u \in Q \text{ and } u' \in P \\ \langle u, v \rangle \leq \langle u', v \rangle\}$$

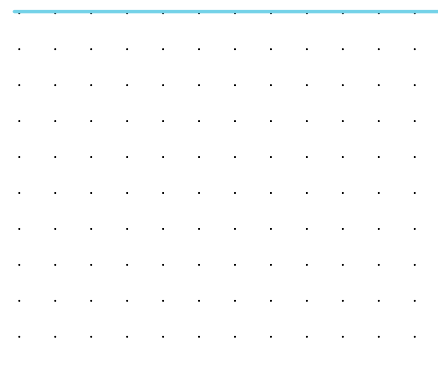
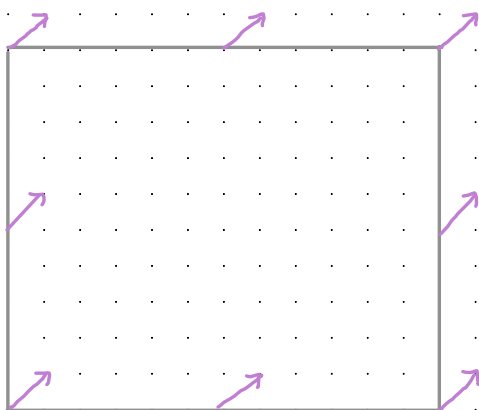
Define

$$\text{Face}_{m,-}(P) = \left\{ C \in \text{Face}(P) \mid \forall c \in C, \text{ the ray } c + \mathbb{R}_{>0} m \text{ does not intersect the interior } P^\circ \text{ of } P \right\}$$

Picture Let $P = [-1, 1]^{x^2} \subset \mathbb{R}^2$ and $m = (1, 0)$



The top, bottom, and right vertical faces are in $\text{Face}_{m,-}(P)$.
 Similarly, if $m = (1, 1)$, then the top and right vertical
 faces are in $\text{Face}_{m,-}(P)$.



Claim 1 The isomorphism ϕ restricts to an isomorphism

$$\text{Face}_{m,-}(P) \xrightarrow{\sim} \Sigma_{m,-}^{\text{op}}$$

Proof of Claim 1.

If $\sigma < \tau$ and $m \notin \sigma^{\vee,0}$, then since $\tau^{\vee,0} \subset \sigma^{\vee,0}$, we also have $m \notin \tau^{\vee,0}$. For $\sigma \in \Sigma_{m,-}^{\text{op}}$, consider the corresponding face C_σ of P . For each $c \in C_\sigma$, the angle spanned by P is the cone $\tau^\vee$ for some $\tau > \sigma$. So the ray $c + \mathbb{R}_{>0} m$ will not pass through P° .

Conversely, let $\sigma \in \Sigma$ be such that $m \in \sigma^{\vee,0}$ (so that $\sigma \notin \Sigma_{m,-}$), and let C_σ be the corresponding face of P . Then at a relative interior point $c \in C_\sigma$, the angle spanned by P is $\sigma^\vee$. Since $m \in \sigma^{\vee,0}$, the ray $c + \mathbb{R}_{>0} m$ passes through P° . $\square$

Claim 2. Let

$$P_{m,-} := |\text{Face}_{m,-}(P)| \quad \text{geometric realization}$$

$$= \bigcup_{Q \in \text{Face}_{m,-}(P)} Q \subset P$$

Then $P_{m,-}$ is contractible.

> If we prove this, we're done. Since $\Sigma_{m,-}^{\text{op}} \simeq \text{Face}_{m,-}(P)$, the classifying anima of $\Sigma_{m,-}^{\text{op}}$ is the anima associated to the topological space $P_{m,-}$.

Proof of claim 2

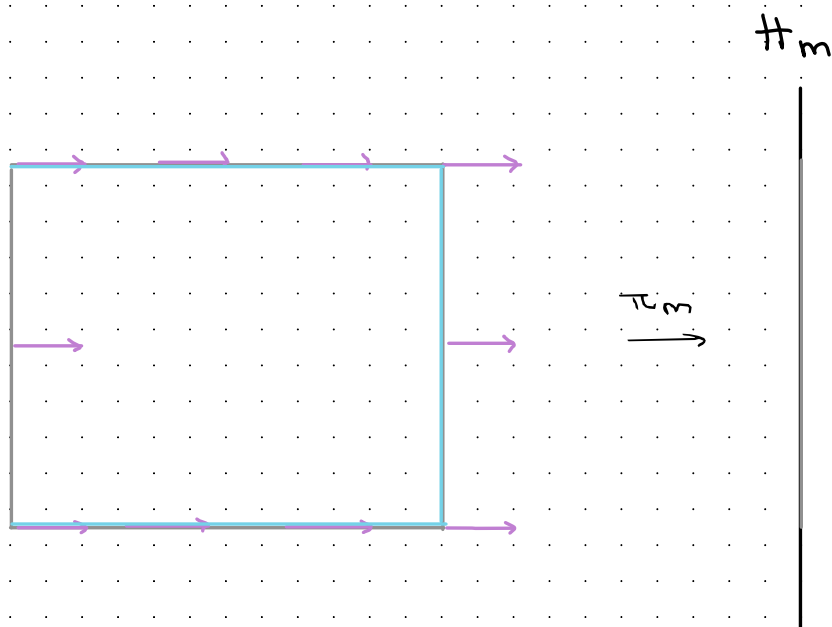
Consider the hyperplane H_m perpendicular to m , and write $\pi_m: M_R \rightarrow H_m$ for the orthogonal projection onto the hyperplane. Observe that

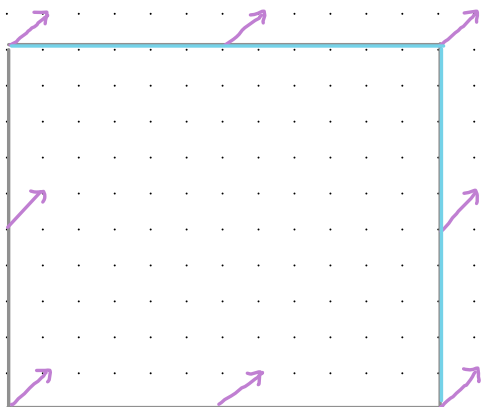
$$\pi_m(P_{m,-}) = \pi_m(P)$$

Moreover, $\pi_m(P)$ is convex, hence contractible. To conclude, check:

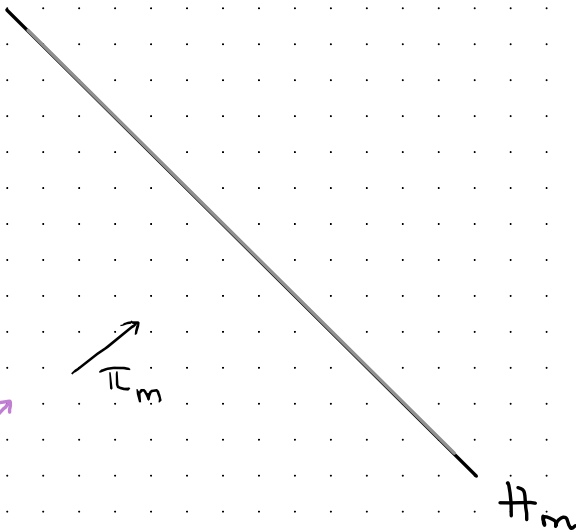
Exercise $\pi_m: P_m, - \longrightarrow \pi_m(P_m, -)$ is a homotopy equivalence.
 Hence $P_m, -$ is contractible. $\square$

> Here are two pictures from the previous examples





π_m



Full faithfulness on generators

Goal Show that for (N, Σ) a smooth fan, the functor

$$\kappa_\Sigma = R_\Sigma \circ \Psi_\Sigma \circ \Phi_\Sigma^{-1}$$

is fully faithful. Since Φ_Σ^{-1} is an equivalence and R_Σ is fully faithful, we equivalently need to show that Ψ_Σ is fully faithful.

> We're aided by the following general result:

Lemma. Let A be an ∞ -category, let $\mathcal{C}_\bullet, \mathcal{D}_\bullet: A \rightarrow \text{Cat}_\infty$ be diagrams, and let $f_\bullet: \mathcal{C}_\bullet \rightarrow \mathcal{D}_\bullet$ be a natural transformation. If for each $\alpha \in A$, the functor f_α is fully faithful, then the induced functor

$$\lim_{\alpha \in A} \mathcal{C}_\alpha \longrightarrow \lim_{\alpha \in A} \mathcal{D}_\alpha$$

is fully faithful.

> So it suffices to show that for each smooth cone $\sigma \in N_{\mathbb{R}}$, the functor

$$\Psi_{\sigma}: \text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}) \longrightarrow \text{Mod}_{\omega_{\sigma}^{\vee}} \text{Sh}(M_{\mathbb{R}})$$

is fully faithful.

- We first check this on generators

Notation For a small ∞ -category $\mathcal{C}_0$, recall that we write

$$Y: \mathcal{C}_0 \hookrightarrow \text{Fun}(\mathcal{C}_0^{\text{op}}, \text{An})$$

for the Yoneda embedding and

$$h: \mathcal{C}_0 \xrightarrow{Y} \text{Fun}(\mathcal{C}_0^{\text{op}}, \text{An}) \xrightarrow{\Sigma_{+}^{\infty, 0, -}} \text{Fun}(\mathcal{C}_0^{\text{op}}, \text{Sp})$$

for the Spectral Yoneda embedding.

Lemma Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ a cone (not nec. smooth). Then the restriction of Ψ_{σ} to the image of the Spectral Yoneda embedding is fully faithful.

> For this, we need

Lemma. Let X be a space and $j: U \hookrightarrow X$ and $j': U' \hookrightarrow X$ opens. Then

$$\mathrm{Map}_{\mathrm{Sh}(X, \mathcal{A}_n)}(j_!(1_{\mathrm{Sh}(U, \mathcal{A}_n)}), j'_!(1_{\mathrm{Sh}(U', \mathcal{A}_n)})) \simeq \begin{cases} \emptyset, & U \not\subset U' \\ *, & U \subset U' \end{cases}$$

and

$$\begin{aligned} \mathrm{Map}_{\mathrm{Sh}(X, \mathcal{S}_p)}(1_U, 1_{U'}) &\simeq \sum_+^\infty \mathrm{Map}_{\mathrm{Sh}(X, \mathcal{A}_n)}(j_!(1_{\mathrm{Sh}(U, \mathcal{A}_n)}), j'_!(1_{\mathrm{Sh}(U', \mathcal{A}_n)})) \\ &\simeq \begin{cases} 0, & U \not\subset U' \\ \mathbb{S}, & U \subset U' \end{cases} \end{aligned}$$

Proof of full faithfulness.

Let $n = \dim(M|_R)$. Recall that for $z \in \Theta(\sigma)$, we have natural equivalences

$$\omega_z \simeq \omega_{z^0} \simeq 1_{z^0}[n]$$

Also observe that the diagram

$$\begin{array}{ccc}
 z & \xrightarrow{\quad} & j_!(1_{\mathrm{Sh}(z, j_{An})}) \\
 \Theta(\sigma) & \xrightarrow{\quad} & \mathrm{Sh}(M_R, An) \\
 \downarrow h & & \downarrow \mathrm{id} \otimes \Sigma_+^\infty \\
 & & \mathrm{Sh}(M_R, Sp) \\
 & & \downarrow s[n] \\
 \mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, Sp) & \xrightarrow{\quad \Psi_\sigma \quad} & \mathrm{Sh}(M_R, Sp)
 \end{array}$$

commutes. Hence the comparison. Hence for $Y, Z \in \Theta(G)$, the induced map

$$\mathrm{Map}(h(Y), h(Z)) \longrightarrow \mathrm{Map}_{\mathrm{Sh}(M_R)}(w_Y, w_Z)$$

is obtained by applying Σ_+^∞ to the natural map

$$\mathrm{Map}_{\mathcal{O}(\mathcal{C})}(Y, Z) \longrightarrow \mathrm{Map}_{\mathrm{Sh}(\mathcal{M}_R, \mathcal{A}_n)}(j_!(1), j'_!(1))$$

where $j: Y^\circ \hookrightarrow \mathcal{M}_R$ and $j': Z^\circ \hookrightarrow \mathcal{M}_R$ are the inclusions.

By the previous lemma, this map is an equivalence. $\square$

Where we're headed

> We'd like to conclude that Ψ_0 is fully faithful by applying the following results.

Note If $\mathcal{C}_0$ is a small ∞ -category and $\mathcal{E} \in \mathbf{Pr}^L$ is compactly gen, then $\mathrm{Fun}(\mathcal{C}_0^{\mathrm{op}}, \mathcal{E})$ is compactly generated.

Lemma Let $\mathcal{C}_0$ be a small ∞ -category, let $\mathcal{D}$ be an ∞ -cat with filtered colimits, and let $f: \mathcal{C}_0 \rightarrow \mathcal{D}$ be a functor. Assume

(1) f is fully faithful.

(2) f factors through the full subcategory $\mathcal{D}^{\mathrm{co}} \subset \mathcal{D}$ spanned by the compact objects.

Then the unique filtered colimit-preserving extension $\mathrm{Ind}(\mathcal{C}_0) \rightarrow \mathcal{D}$ of f is fully faithful.

Two problems

(1) We know that the image of the spectral Yoneda embedding in $\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp})$ consists of compact objects. However they don't generate $\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp})$ under filtered colimits.

(2) As stated earlier, $\text{Sh}(M_{\mathbb{R}})$ has no nonzero compact objects

> Problem (1) is easy to resolve with some elementary category theory

> Problem (2) is more serious. The solution is to find a full subcategory of $\text{Mod}_{\omega_{\sigma^v}} \text{Sh}(M_{\mathbb{R}})$ through which Ψ_{σ} factors that is compactly generated and closed under colimits. Then we can try to check that Ψ_{σ} factors through the compact objects of this subcategory.

- Enter: constructible sheaves and exit-paths.